

SPECIAL REPORT

ASPEN definitions in pediatric intestinal failure

Biren P. Modi MD, MPH¹  | David P. Galloway MD²  | Kathleen Gura PharmD³  |
Anita Nucci PhD, RD⁴  | Steven Plogsted PharmD⁵ | Alyssa Tucker MS, RD⁶ |
Paul W. Wales MD, MSc On behalf of the Pediatric Intestinal Failure Section of ASPEN⁷

¹ Center for Advanced Intestinal Rehabilitation, Department of Surgery, Boston Children's Hospital and Harvard Medical School, Boston, Massachusetts, USA

² Division of Gastroenterology, Hepatology and Nutrition, University of Alabama at Birmingham, Birmingham, Alabama, USA

³ Division of Gastroenterology, Hepatology and Nutrition, Boston Children's Hospital and Harvard Medical School, Boston, Massachusetts, USA

⁴ Department of Nutrition, Georgia State University, Atlanta, Georgia, USA

⁵ , Columbus, Ohio, USA

⁶ Department of Clinical Nutrition, Children's National Medical Center, Washington, DC, USA

⁷ Group for Improvement of Intestinal Function and Treatment, Department of Surgery, Hospital for Sick Children and University of Toronto, Toronto, Ontario, Canada

Correspondence

Biren P. Modi, MD, MPH, Center for Advanced Intestinal Rehabilitation, Department of Surgery, Boston Children's Hospital, Fegan 3, 300 Longwood Ave, Boston, MA 02115, USA.
Email: biren.modi@childrens.harvard.edu

Abstract

Pediatric intestinal failure (PIF) is a relatively rare disease entity that requires focused interdisciplinary care and specialized nutrition management. There has long been a lack of consensus in the definition of key terms related to PIF because of its rarity and a plethora of small studies rather than large trials. As such, the American Society for Parenteral and Enteral Nutrition (ASPEN) PIF Section, composed of clinicians from a variety of disciplines caring for children with intestinal failure, is uniquely poised to provide insight into this definition void. This document is the product of an effort by the Section to create evidence-based consensus definitions, with the goal of allowing for appropriate comparisons between clinical studies and measurement of long-term patient outcomes. This paper has been approved by the ASPEN Board of Directors.

KEYWORDS

definition, enteral nutrition, intestinal failure, parenteral nutrition, pediatrics, short bowel syndrome

CLINICAL RELEVANCY STATEMENT

This manuscript provides evidence-based definitions for key terms in the field of pediatric intestinal failure. Given a lack of consensus definitions and a variability in definitions utilized in scientific endeavors, a comparison of outcomes and the ability to accurately determine best practices have been limited. The American Society for Parenteral and Enteral Nutrition (ASPEN) Pediatric Intestinal Failure Section is uniquely situated to develop these necessary definitions by providing input from stakeholders within a variety of healthcare professions and through access to the most up-to-date research and expert opinion. The definitions provided in this manuscript should have direct rele-

vance to the daily management of children with intestinal failure by allowing all providers, across disciplines and at different centers, to speak a common language.

INTRODUCTION

The American Society for Parenteral and Enteral Nutrition (ASPEN) Pediatric Intestinal Failure (PIF) Section consists of dietitians, nurses, pharmacists, physicians, and other affiliated clinicians who participate in the management of children with long-term complex nutrition needs secondary to intestinal dysfunction. As members of a broader PIF community, this group recognized an inconsistency in the reporting and

TABLE 1 American Society for Parenteral and Enteral Nutrition definitions of seven key terms associated with pediatric intestinal failure

Term	Definition
Pediatric intestinal failure	Pediatric intestinal failure is the reduction of functional intestinal mass below that which can sustain life, resulting in dependence on supplemental parenteral support for a minimum of 60 days within a 74 consecutive day interval.
Parenteral support	Parenteral support involves provision of IV fluids in those patients who are unable to maintain full enteral autonomy. These fluids can contain any combination of macronutrients, electrolytes, vitamins, or trace elements. PN, as a subtype of parenteral support, is the IV provision of nutrients in those patients who are unable to adequately supply nutrients to their body by enteral means in order to treat or prevent the development of disease-related malnutrition. It may contain protein, carbohydrate, and/or fat, as well as electrolytes, trace elements, micronutrients, and/or vitamins along with an adequate amount of fluid for hydration, and may provide partial or full estimated needs.
Enteral support	Enteral support involves the provision of sufficient nutrition and fluids through the intestinal tract to facilitate appropriate growth and hydration with or without parenteral support or nutrition. These fluids can contain any combination of macronutrients, micronutrients, or electrolytes. Enteral nutrition, as a subset of enteral support, is the specific provision of macronutrients through the intestinal tract. It may contain protein, carbohydrate, and/or lipid, along with an adequate amount of fluid for hydration, and may provide partial or full estimated needs. The goal of enteral support is to promote intestinal adaptation, enhance the ability to wean PN, aid in the prevention of cholestasis, limit morbidity and mortality by promoting enteral autonomy, and promote developmentally appropriate oral feeding skills.
Enteral autonomy	Enteral autonomy is the maintenance of normal growth and hydration status by means of enteral support without the use of parenteral support for a period of >3 consecutive months.
SIBO	SIBO is objectively defined as • an excess of bacteria $>10^5$ CFU/ml in duodenojejunal aspirate fluid ($>10^3$ CFU/ml if species identified are normally present in the colon; that is, gram negatives, enterococci, anaerobes), OR • a glucose or lactose hydrogen breath test with either double peak or a peak of >20 parts per million hydrogen above basal within 90 min of lactulose ingestion. In the absence of objective data, symptoms such as abdominal pain, bloating, diarrhea, or flatulence that respond to a course of empiric antibiotic therapy can be used.
IFALD	IFALD describes liver injury, as manifested by cholestasis, steatosis, and fibrosis, in patients with intestinal failure that is independent of, or in addition to, other potential etiologies. The development of IFALD is multifactorial, typically as a consequence of metabolic abnormalities in intestinal failure and the medical and surgical management strategies of intestinal failure themselves. It can be stabilized or reversed with appropriate early modification of management strategies and promotion of intestinal adaptation or it can progress to hepatic dysfunction and end-stage liver disease.
IRP	An IRP is an interdisciplinary, collaborative patient care paradigm that serves to coordinate care for children with intestinal failure through comprehensive management of their specialized nutrition and corollary needs, attention to and support for associated chronic comorbidities, and evaluation and treatment of acute complications.

Abbreviations: CFU, colony-forming unit; IFALD, intestinal failure–associated liver disease; IRP, intestinal rehabilitation program; IV, intravenous; PN, parenteral nutrition; SIBO, small-intestinal bacterial overgrowth.

study of PIF resulting from a lack of uniform definitions for key terminology in the field. As such, the Section charged a working group of representatives from the respective disciplines to identify and define key terms whose standardization would provide clarity to clinical and research reporting and allow for comparison of results, longitudinal evaluation of management methods, and alignment of outcomes metrics across studies and centers managing PIF. The final definitions of the chosen terms are listed in Table 1, with a more detailed accounting of each term's definition derivation given in the document.

Any recommendations in this paper do not constitute medical or other professional advice and should not be taken as such. To the extent that the information published herein may be used to assist in the care of patients, this is the result of the sole professional judgment of the attending healthcare professional whose judgment is the primary component of quality medical care. The information presented here is

not a substitute for the exercise of such judgment by the healthcare professional. Circumstances in clinical settings and patient indications may require actions different from those recommended in this document, and in those cases, the judgment of the treating professional should prevail. This paper has been approved by the ASPEN Board of Directors.

METHODS

At the annual meeting of the PIF Section during the 2018 ASPEN Nutrition Science & Practice Conference, the Section chair brought forward a proposal to address the need for uniform definitions focused on the nutrition management of patients with PIF. The Section agreed to create a working group representative of the interdisciplinary care of

these patients, which would be headed by the chair and tasked with identifying the key terms that most needed standardized definitions, creating a rubric for identifying definitions available in the literature and through expert opinion, and distilling the salient points into a single standardized definition for each term.

The group was formed via member nomination, with all members of the PIF Section eligible to participate and allowing for representation from multiple disciplines and across multiple institutions. The group was convened via conference call in March 2018. The work of the group was carried out from March 2018 to March 2019 through monthly conference calls or meetings. The group began by identifying key terms in PIF and then selecting from these the terms that would be most impactful by having standardized definitions. The final list consisted of the seven terms included in this document, along with several subordinate terms that were felt to be of sufficient import that they should be specifically addressed. The final list included “pediatric intestinal failure,” “parenteral support,” “enteral support,” “enteral autonomy,” “small-intestinal bacterial overgrowth,” “intestinal failure–associated liver disease,” and “intestinal rehabilitation program.” Once the final list of terms was determined, each term was assigned to a member of the group to perform a Medline search limited to English-language articles. The results of the search were summarized by the member, presented to the group, and then narrowed to nonredundant articles that had specific definitions proposed for the term. Additional articles were recommended by members of the working group, and additional components of the term’s definition were proposed by the members, constituting expert opinion. Delphi methodology was used to then form and finalize the definition over the course of three to five conference calls for each term.^{1,2} Each Delphi round consisted of a definition proposal by the term’s member sponsor, feedback and comments by the working group, additional Medline search if necessary, and then definition refinement. Term definitions were considered final once there was 100% consensus among the working group members to approve the definition.

The definitions were finalized and presented to the ASPEN PIF Section for approval at the 2019 ASPEN Nutrition Science & Practice Conference. The proposed definitions were circulated prior to the in-person meeting, and members who would not be present at the meeting in Phoenix, Arizona were allowed to provide feedback via email to the Section chair. Using this methodology, the group created a predetermined requirement that 80% approval would be required from the members of the Section for finalization of each definition. Ultimately, all definitions were approved as submitted by the working group with suggestions regarding discussion points for manuscript preparation provided by the Section members, but no rejection of the definitions themselves.

After final approval, the definitions document was prepared with each member composing the Section relevant to their term, providing the definition as approved by the Section and the rationale and supporting literature as well as areas of controversy or future study. This manuscript was then approved by the ASPEN Board of Directors.

DEFINITIONS

Pediatric intestinal failure

PIF is the reduction of functional intestinal mass below that which can sustain life, resulting in dependence on supplemental parenteral support for a minimum of 60 days within an interval of 74 consecutive days.

Background

For centuries, children of all ages have been the unfortunate recipients of either congenital or acquired intestinal conditions that resulted in severe bowel dysfunction or significant intestinal loss. In most cases, these events would render the patient incapable of the absorption of fluid and nutrients essential for life, which would ultimately lead to death. This was viewed as a “failure of the intestine” to function properly. By the late 1960s, certain developments, such as the parenteral delivery of fluid and nutrients, were being implemented in an attempt to prolong life in the face of conditions that were leading to a failure of intestinal function in children.

As the development of parenteral nutrition (PN) continued, so did the survival of many infants and children who were victims of intestinal insults from diagnoses such as necrotizing enterocolitis, complicated gastroschisis, and chronic intestinal pseudo-obstruction. The term “intestinal failure” (IF) was used more often to identify children with some degree of reliance on PN. Decades later, IF now encompasses more than just a medical condition. It involves communities of patients, families, and care providers working together in partnerships and interdisciplinary teams. Significant research is ongoing to move the field forward. As the field continues to grow, so does its complexity, which has led to the current need for definitions that will provide consistency and agreement on the terms used to describe IF in both research and clinical practice.

Definition

A PubMed search performed for the term “intestinal failure,” with filters (review, humans, English, Child: birth–18 years), resulted in 445 articles for review. A majority of the articles retrieved ($n = 430$, 96%) were excluded because they failed to provide an overview of IF and/or did not include a definition of IF. Of the remaining 15 articles, 10 mentioned a component of “functional intestinal mass” in their definitions.^{3–11} Three articles specified a “dependence on PN,”^{4,12,13} and only one article stipulated length of time receiving PN in days as part of the definition.¹³

A separate PubMed search was also performed for the term “short bowel syndrome” (SBS), with filters (review, humans, English, Child: birth–18 years). This resulted in 243 articles for review. Articles that did not provide a definition for SBS were excluded ($n = 216$, 88%). Of the remaining 27 articles, more than half ($n = 15$, 55%) specify the small

bowel being involved in the etiology of SBS.^{12,14–27} An emphasis on a functional definition of SBS was observed in eight articles,^{20,23,24,27–32} and only one article included a specific residual bowel length in the definition.³³

A subsequent PubMed search was performed for the term “ultrashort bowel syndrome,” with filters (humans, English, Child: birth–18 years). This yielded 30 articles for review, of which only a minority ($n = 7$, 23%) were included. Excluded articles either did not pertain to SBS or did not propose a definition for ultrashort-bowel syndrome. Among those articles reviewed, the majority propose specific residual small-bowel lengths in centimeters (cm) to define ultrashort-bowel syndrome.^{14,29,34–37} Small-bowel lengths ranged from 10 to 25 cm. In addition to length of residual small bowel in centimeters, another article proposes a residual proportional length cutoff of 10% expected small bowel.¹⁴ Finally, one article makes no mention of residual small-bowel length but rather emphasizes a potentially irreversible dependence on PN as part of the definition for ultrashort-bowel syndrome.³⁸

The PIF definitions working group agreed that the definition of PIF should be inclusive of all pediatric patients that are unable to sustain energy/volume needs without the use of parenteral support. Parenteral support needs may be partial or complete, realizing that, depending on anatomy, some patients require a greater percentage of their energy needs parenterally than others. The definition does not state specifically a residual bowel length that can result in IF, because IF, defined broadly, relies on a functional deficit of intestinal mass. In this regard, this definition is inclusive of all disorders of the bowel, which include SBS and chronic intestinal pseudo-obstruction. A definition based solely on bowel length would exclude many conditions that require intestinal rehabilitation and/or complex nutrition management. With respect to duration of PN, the working group agreed to include a minimum number of days of PN dependence in the definition. This is consistent with the definition proposed by the Pediatric Intestinal Failure Consortium (PIFCON) and enables proper identification of patients with prolonged PN needs and IF.¹³

Subordinate definitions

SBS is the development of IF secondary to loss of small bowel, either from congenital absence or following surgical resection.

The working group agreed to include “short-bowel syndrome” as a subcategory of IF. It was decided that the definition for SBS should first meet the requirements previously stated in our definition for IF but specifies the cause of IF is overall loss of small-bowel length, whether due to resection or due to a congenital absence. The length of resected small bowel that results in SBS is purposefully not included, given that there is a large variation in residual small-bowel lengths that result in the need for prolonged parenteral support.

Ultrashort-bowel syndrome is the development of IF following significant small-bowel resection resulting in a residual small-bowel length that is $<10\%$ of expected.

Finally, the working group also approved the inclusion of “ultrashort-bowel syndrome” as a subcategory of IF and SBS. The

working group agreed that a definition of ultrashort-bowel syndrome should first meet the requirements of having both IF and SBS, with the additional reference to percentage of remaining residual small-bowel length, most commonly calculated using the method of Struijs et al.³⁹ The working group unanimously felt that these patients do exhibit a different phenotype in regards to enteral tolerance and infection risk, which justifies the creation of this subcategory of SBS.

Controversies and future directions

IF has been defined in many different ways in the literature over the past few decades. Some definitions specify dependence on parenteral support, whereas others make no such reference. Although most definitions historically have not included a specific time period of dependence on PN, such a statement is important to try and capture patients with true IF who will require intestinal rehabilitation. A more concise definition will allow proper identification of patients with IF and facilitates more-consistent research from which the community can continue to learn about this common medical condition.

With respect to SBS, the definition has been more unclear historically. Some definitions have emphasized the small bowel as being primarily responsible for SBS, whereas others mention “bowel” or “intestine” only, leaving room for speculation. Although the role of the colon in intestinal rehabilitation has been well documented, SBS principally results from the loss of small bowel (surgical or congenital), the primary site for absorption of nutrients. Another point of discussion is whether SBS should be defined by a particular residual small-bowel length or strictly by loss of function. It is essential to have some degree of small bowel remaining to adequately absorb enteral delivery of fluid and nutrients. However, there are many factors that influence the ability for a patient with IF to wean off PN other than small-bowel length alone,³⁸ which may explain why two participants with the same residual small-bowel length may differ dramatically in their ability to achieve enteral autonomy. Therefore, emphasis on remaining intestinal function, rather than a threshold residual small-bowel length, is made in the proposed definition. Further studies will continue to illustrate the importance of the functional capacity of residual small and large bowel in the management of SBS.

Over the last decade, ultrashort-bowel syndrome has begun to garner some attention in the literature. However, as a separate entity and subcategory of SBS, it has not been well defined. Although absolute length in centimeters has been more often cited, the working group believes residual proportional small-bowel length will be more meaningful to describe these patients in the future. Additionally, the working group thinks that patients with ultrashort-bowel syndrome, as defined here, likely have a different risk profile when it comes to sequelae of their IF. Once identified, patients with ultrashort-bowel syndrome should not be labeled as unable to wean from PN. Rather, these patients still require and merit rehabilitation. This expert opinion justifies the need for more attention to be placed on this subset of patients, not only in clinical practice but also through additional research efforts.

Parenteral support

Parenteral support involves the provision of intravenous (IV) fluids in those patients who are unable to maintain full enteral autonomy. These fluids can contain any combination of macronutrients, electrolytes, vitamins, or trace elements. PN, as a subtype of parenteral support, is the IV provision of nutrients in those patients who are unable to adequately supply nutrients to their body by enteral means in order to treat or prevent the development of disease-related malnutrition. It may contain protein, carbohydrate, and/or fat, as well as electrolytes, trace elements, micronutrients, and/or vitamins, along with an adequate amount of fluid for hydration, and may provide partial or full estimated needs.

Background

Parenteral support and/or PN is required to provide full or partial nutrients needed to adequately maintain or promote appropriate growth and body composition in the absence of full enteral autonomy. Robert Elman, in the late 1930s, observed that protein hydrolysates could safely be administered in humans, and in 1961, Dr Arvid Wretling developed soybean oil-based IV lipid. This was followed by the groundbreaking work of Dr Stanley Dudrick demonstrating that a catheter placed into a central vein could deliver nutrients safely for an extended period.⁴⁰ Since that time, progress has been made to improve the outcome of patients who could not depend on survival strictly by enteral means. With the increase in methods to deliver PN and the desire to provide this lifesaving therapy came some unwanted effects. Metabolic and physiological derangements such as electrolyte abnormalities, glucose intolerance, and liver damage became known. These findings provided incentives to develop the optimal way to deliver parenteral therapy to maximize growth and minimize the potential harm to the patient. In addition, the importance of nonnutrient parenteral therapies has also become evident. During the second worldwide cholera outbreak, in 1832, Dr Thomas Latta demonstrated the benefit of injecting a salt solution into the veins for the purpose of improving recovery of dehydrated cholera patients.⁴¹ Vitamin use is of critical importance in providing parenteral therapy. Although ASPEN has published guidelines on appropriate vitamin mixtures,⁴² these may not fully meet the demands of patients with IF, and vigilance must be maintained to prevent nutrition deficiencies even in patients with IF who are receiving parenteral therapy.⁴³

Definition

A PubMed search performed for the terms “parenteral support” and “nutrition,” with filters (review, humans, English), resulted in 8868 articles for review. Most articles retrieved ($n = 8607$, 97%) were excluded because they failed to provide an overview of parenteral support/nutrition and/or did not include a novel definition. Of the remaining 61 articles, the majority gave a general reference to “par-

enteral support” and “nutrition”; however, several of the articles attempted to define the terms in greater detail and are summarized in this article.^{44,45}

Of the remaining 61 articles, only four articles discussed a dependence on PN as part of the definition.^{6,46–48}

The goals of parenteral support/nutrition are to provide full or partial nutrients needed to adequately maintain or promote appropriate growth and body composition in the absence of full enteral autonomy.^{44,45} There are several items that fall under this umbrella, and they include IV amino acids/dextrose admixtures, with or without intravenous lipid emulsion (ILE); ILE with or without other fluids; IV fluids with a dextrose concentration $>5\%$; IV fluids containing trace elements; and IV fluids containing vitamins.

Parenteral support involves providing basic fluids in those patients who are unable to attain full enteral autonomy. These fluids can contain any combination of micronutrients such as electrolytes, vitamins, or trace elements.^{44,45} PN is an IV form of nutrition used by patients who are unable to adequately supply nutrients to their body by enteral means in order to treat or prevent the development of disease-related malnutrition.^{6,48} It may be provided either through a central or peripheral vascular access devices. It may contain protein, carbohydrate, fat, or micronutrients such as electrolytes, trace elements, and vitamins, along with an adequate amount of fluids for hydration, and may provide partial or full estimated nutrition needs.

Controversies and future directions

Parenteral support and PN are described in the literature in general terms; however, no single definition has been formulated. The definition provided here is a product of the small amount of information available in the literature and clinically practical points obtained through the Delphi process conducted by content experts. Future research may lead to a standard definition, but for now, an acceptable consensus definition has been put forward. Additional research may also be needed to identify the optimal components of such parenteral support for each patient.

Enteral support

Enteral support involves the provision of sufficient nutrition and fluids through the intestinal tract to facilitate appropriate growth and hydration with or without parenteral support or nutrition. These fluids can contain any combination of macronutrients, micronutrients, or electrolytes. Enteral nutrition (EN), as a subset of enteral support, is the specific provision of macronutrients through the intestinal tract. It may contain protein, carbohydrate, and/or lipid, along with an adequate amount of fluid for hydration, and may provide partial or full estimated needs. The goal of enteral support is to promote intestinal adaptation, enhance the ability to wean PN, aid in the prevention of cholestasis, limit morbidity and mortality by promoting enteral autonomy, and promote developmentally appropriate oral feeding skills.

Background

Over recent decades, research has grown not only in the area of IF but also in how we deliver nutrients to the bowel, what we put in the bowel, and for what purpose. This has led to the meaning of “enteral support” to become multifactorial. The complexity of IF requires an individualized approach with EN. Therefore, it is important to define clear parameters of what enteral support entails to provide guidance for clinical practice, as well as future research. It is also important to note that enteral support differs from enteral autonomy, which is later defined.

Definition

A PubMed search was performed for the terms “enteral support short bowel,” using the standard English language and human participant filters from January 1970 to March 2019. The search resulted in 107 articles that were reviewed, 12 of which were ultimately included. To minimize duplicates, only articles with “original” definitions and classifications were chosen. In addition to this limited reference material available, the definitions working group as well as the members of the PIF Section were solicited for expert opinion on the matter.

Enteral support includes the maintenance of fluid and electrolyte balances with sufficient nutrition to facilitate growth and maximize intestinal adaptation while routinely monitoring vitamin and mineral levels.^{34,49,50} Enteral support includes nutrition delivered to the gastrointestinal (GI) tract either by mouth or enteral access device (EAD), such as nasogastric tubes, nasogastrojejunal tubes, and percutaneous endoscopic gastrostomy and jejunostomy, or as determined by medical provider. It includes initiating and advancing EN, in combination with strategies of determining the appropriate type, route, and rate of feed; additive/modular components; and vitamin and mineral supplementation. The goal of enteral support is to promote intestinal adaptation by stimulating mucosal hyperplasia, GI hormone secretion, and pancreaticobiliary secretions, with the ultimate goal being enteral autonomy.⁵⁰ No one factor is attributed to the success of achieving enteral autonomy though enteral support. Enteral support should be individualized and based on the presence or absence of colon, the presence or absence of the ileocecal valve, remnant bowel anatomy, and tolerance.

Enteral support may be initiated anytime after surgery, typically ranging from 5 days to 2 weeks.⁵⁰ Continuous EN is the preferred method over bolus administration secondary to reduced intolerance, promotion of adaptation via continual saturation of the intestinal lumen, and decreased risk of osmotic diarrhea; however, there is limited research evidence supporting this method.^{49–51} Bolus administration is more physiologic, mimicking the typical infant feeding pattern and allowing for periods of fasting, possibly preventing hyperinsulinemia and promoting appropriate bacterial clearance.⁵⁰ If a patient is developmentally and clinically able to initiate oral feeds, small-volume oral boluses are encouraged to reduce oral aversion.⁵⁰ Criteria for the initiation of oral feedings need to be established. The timing of initiation of oral feedings has yet to be determined, and oral motor therapy is important to the intestinal rehabilitation process.⁵⁰ If possible, a com-

bination of oral feedings (to maintain appropriate oral skills and promote normalization) and continuous EN (to promote intestinal adaptation and tolerance) is recommended. The rate at which EN is advanced should be individualized with consideration of volume stool/ostomy output, consistency of stool output, vomiting, and/or abdominal distention in association with institution-based feeding guidelines.^{49,51,52} Goal stool output should be limited to 40–50 ml/kg/day.⁴⁹ Postsurgical feeding advancement protocols should be implemented to aid in the standardization of care and improve patient outcomes.

Recent studies evaluating postoperative feeding guidelines have found that implementation of such guidelines reduce the risk of IF-associated liver disease (IFALD).^{52,53} The definitions working group encourages the use of institution-specific standardized feeding protocols in the postsurgical neonates. Such guidelines aid in consistent communication among disciplines, compliance among clinicians, improved patient outcomes, and consistency in further research.^{52,53}

Type of feeding

Human breast milk is the first choice of feedings, and amino acid–based formula, if breast milk is unavailable, has demonstrated a shorter duration of PN.^{51,54} This could be attributed to human breast milk containing long-chain fatty acids, free amino acids, and growth factors that, when paired with immune-enhancing properties, may assist in intestinal adaptation.^{50,51} A high percentage of long-chain fatty acids, which may promote mucosal adaptation, in combination with medium-chain triglycerides, which are directly absorbed by enterocytes, supports amino acid–based formulas as the formula of choice for most centers.⁵⁰ If human breast milk is unavailable, clinical judgment is recommended, accounting for injury to bowel, remaining intestinal length, gestational age when selecting a feeding alternative, and concern for milk protein allergy.

Additives and modular EN products

There are numerous ways a clinician can manipulate EN to optimize growth and/or feeding tolerance while encouraging enteral autonomy. This may include the addition of soluble fiber, fat modulars, and/or vitamin and mineral supplementation. Soluble fiber has been shown to be beneficial because it is fermented into short-chain fatty acids, which can be metabolized by colonocytes for energy, decrease transit time, and add bulk to stools.^{50,55} Various forms of fiber, such as the fiber in green beans, wheat dextrin, or guar gum, have been reported to treat high-output and/or loose stools.^{50,56} In a recent survey, 67% of clinicians used pectin, a water-soluble fiber, as a treatment for increased stool output.⁵⁶ Fat modulars in the form of long-chain or short-chain triglycerides may be used, as they provide minimal change in the osmolality while providing additional energy.

Vitamin and mineral supplements are often added to feeding regimens. Resected portions of the bowel should be taken into account when assessing vitamin and mineral intake, as well as routine

monitoring of vitamin/mineral status.⁷ A recent study reported that during the transition from PN to EN over a 12-week period, 33% of patients had one vitamin deficiency, with 77% having one mineral deficiency.⁵⁷ Once the patients transitioned to full EN, vitamin deficiency was reported to increase up to 70%, with no change in mineral deficiency.⁵⁷ Iron was also noted to be the most common deficiency, with >90% of the patients being anemic during the transition to full EN, followed by zinc and vitamin D.^{34,51,58} A multivitamin should routinely be prescribed for these patients, in addition to specific mineral supplementation in children with SBS/IF.⁵⁸ Vitamin and mineral levels should be monitored routinely, especially during the transition from PN to EN, and continued when the patient reaches enteral autonomy.³⁴

Some patients with PIF related to SBS require additional fluids to maintain adequate hydration and/or electrolyte balance, which may be in the form of oral rehydration solutions (ORSs). Because of its purpose and its inherent delivery into the GI tract, we categorize ORS as providing enteral support. There is much variability in the use of these additives/modular EN products, as well as vitamin and mineral supplementation. Their use should be individualized.

The introduction of solid foods should occur when developmentally appropriate, generally around 4–6 months of age. Children with PIF may be developmentally delayed or experience oral aversion.^{51,59} Clinicians should encourage, as medically appropriate, the intake of nutrient- and energy-dense solid food intake and provide guidance to patients on what that includes.⁵¹ Because of the increased risk of oral aversion in this population, often times complementary solid foods may not add to energy intake but are used for developing and maintaining oral feeding skills. The effect of certain foods and quantity should be closely monitored, as stool output and/or tolerance may be affected. There is also growing popularity in the use of “real foods” or blended solid foods used through an EAD, instead of manufactured structured formula. This topic is growing in popularity, with minimal evidence, and should continue to be studied, especially considering many of these products are not nutritionally complete. Some commercial products are currently available and marketed as “blended diets,” and therefore, we consider them to fall under the definition of enteral support.

EN ordering, preparation, labeling, and delivery, for patients in both the inpatient and outpatient setting, may be complex. We recommend referring to “ASPEN Safe Practices for Enteral Nutrition Therapy” for guidance to minimize the potential risk of errors and ensuring safe delivery of enteral support, especially to this high-risk population.⁶⁰

Controversies and future directions

The working group discussed whether to include oral intake as part of enteral support, as it varies within the literature. The group concluded that enteral support included anything that was fed into the GI tract, especially because there is such a variety in this population of nutrient delivery routes. We did not want to exclude patients who may rely on oral supplements (eg, formula or ORS) or receive a combination of nutrition through an EAD and oral nutrition. Therefore, for the purpose of defining enteral support, as it pertains to SBS/IF, enteral sup-

port includes anything fed into the GI tract orally and/or through an EAD.

There are numerous opportunities for future research with enteral support. Future studies should continue to examine timing and advancement of feedings (at all age groups), what is best practice for feeding (eg, formula, breast milk, and blenderized diets), additives to EN (eg, fiber), the use of ORS, and developmentally appropriate feeding methods within this population. Multicenter studies are warranted to find the best practices for the variety of therapeutic strategies within enteral support.

Enteral autonomy

Enteral autonomy is the maintenance of normal growth and hydration status by means of enteral support without the use of parenteral support for a period of >3 consecutive months.

Background

A primary management strategy for children with SBS is to reduce the concentration of nutrients provided by PN while enteral nutrient intake is increased, with the goal of achieving enteral autonomy. Maintenance of nutrition status and growth in children with SBS can be challenging because of the clinical manifestations of the disorder, which include feeding intolerance, altered intestinal motility, malabsorption of electrolytes and macronutrients and micronutrients, and oral aversion.^{61–63} After children with SBS are stabilized postoperatively, standard practice has been to gradually cycle PN administration hours downward while concurrently maximizing EN or oral feedings. Despite the use of PN and standard medical management, growth failure by anthropometry is still observed in a high percentage of patients. Rates of underweight (weight for age less than the fifth percentile) and stunting (length or height for age less than the fifth percentile) have been reported between 21% and 38% and 34% and 46%, respectively, in this population.^{13,27}

Children with IF are at risk for multiple complications, including metabolic abnormalities, mechanical and infectious complications of vascular access devices, structural and functional bowel disorders, chronic liver disease, and a lower quality of life.¹³ Morbidity and mortality rates in PIF have been associated with many factors, including (1) age at the time of surgery, (2) residual bowel length, (3) function and adaptive capacity of the remnant bowel, (4) the ability to achieve enteral autonomy, (5) incidence of sepsis, and (6) development of IFALD.^{55,61} The most effective strategy for stimulating intestinal adaptation, achieving intestinal rehabilitation, and reducing the risk of PN-related complications remains the provision of EN.⁵⁵ Achieving enteral autonomy is thus a primary clinical management goal as well as outcome measure to evaluate the effectiveness of treatment strategies. Therefore, a standard definition of enteral autonomy is important for both clinicians and researchers to allow for determination of best practices to achieve optimal outcomes.

Definition

A PubMed search performed for the terms “enteral autonomy” and “review” (publication type), with filters (humans, English), resulted in 57 articles that were reviewed for a definition of enteral autonomy. An additional seven articles retrieved from the reference lists of the initial article list were also reviewed. The vast majority of articles ($n = 56$, 87%) did not include a definition of enteral autonomy. Of the eight articles (two review papers and six retrospective cohort studies) that did include a specific or implied definition of the term, all included the discontinuation of PN.^{54,64–70} The authors of three articles provided a qualifying duration of discontinuation (>3 consecutive months⁶⁷; ≥ 6 months⁶⁵; ≥ 12 months⁵⁴). In one article, the definition of enteral autonomy included the maintenance of adequate growth,⁶⁷ whereas in two others, the definition included the maintenance of adequate growth and fluid and electrolyte balance.^{54,68} One article included the discontinuation of enteral feeding devices in addition to vascular access devices.⁶⁹ Based on these references, the definitions working group determined that the definition of enteral autonomy should include mention of a duration of discontinuation of PN as well as a measure of normal growth and hydration status. None of the articles that included a definition of enteral autonomy mentioned the impact of brief resumption of parenteral support due to acute illness. In addition, the need for enteral or parenteral micronutrient supplementation was not mentioned as a disqualifier for achievement of enteral autonomy. The working group members agree that brief resumptions of parenteral support (nutrition and/or hydration) for up to 2 weeks after enteral autonomy is achieved or the need for enteral/intramuscular vitamin or micronutrient supplementation would not disqualify a patient from enteral autonomy status.

A second PubMed search for the terms “normal growth” or “adequate growth,” “SBS” or “intestinal failure,” and “review” (publication type), with filters (humans, English), was conducted to identify a definition for normal growth in the PIF population. Of the resulting 484 articles that were reviewed, only five (2%) included a specific or implied definition of the term. The authors of two articles defined normal growth as reaching predicted height for age,^{71,72} two articles referred to reaching or exceeding a specific percentile on the growth curve (normal growth is ≥ 25 th percentile for weight, length, and weight for length⁷³; delayed growth less than the fifth percentile⁷⁴), and one article defined children with chronic growth failure as having z-scores for length more than two standard deviations below the mean (less than the third percentile).⁷⁴ Some children with IF have anthropometric parameters that never reach the growth curve. Therefore, assessment of growth status cannot always be determined by using percentiles. Mehta et al recommended using a decline in z-score for individual anthropometric measurements as a measure of faltering growth.⁷⁵ The period between measurements will vary, and accommodation needs to be made for measurement error.

Communication with the ASPEN PIF Section dietitian members about the definition of normal growth resulted in discussion about

the importance of adequate nutrition in the assessment of growth status. Therefore, the working group agreed upon the following definition of normal growth to be used when determining adequacy of enteral autonomy: sustained maintenance (eg, >3 months) of a minimum length-for-age or height-for-age z-score (eg, a decrease of no more than 0.5 standard deviation) from baseline steady state or time point determined by the healthcare professional or researcher while meeting estimated nutrient requirements.

Controversies and future directions

Enteral autonomy has been used as a primary outcome measure in multiple studies related to the management of PIF. Despite the importance of achieving enteral autonomy in the reduction of complications, intestinal adaptation, and quality of life, the term has not been well defined. Definitions of enteral autonomy vary not only in the time off of PN but also in the degree to which the child is autonomous (with or without IV hydration, enteral feeding devices, and the ability to maintain normal growth). Hopefully, with this strict definition of enteral autonomy, it will be possible to determine management strategies that result in the best outcome for children with IF. Similarly, the definition of maintenance of hydration status has not yet been determined and is a potential target of future work to allow for more uniformity.

Children who successfully transition from PN to EN are also at risk for micronutrient deficiencies.^{57,58} Those who achieve enteral autonomy after autologous intestinal reconstruction surgery and subsequently experience small-bowel redilation have been found to be at risk for resumption of PN.^{76,77} The working group recommends regular monitoring of micronutrient and growth status as well as intestinal function after transition to full EN. Future studies should examine growth status in the short and long term after PN weaning. In addition, the use of alternative methods of growth assessment, such as body composition measures and other anthropometric measures, should be evaluated.

Small-intestinal bacterial overgrowth

Small-intestinal bacterial overgrowth (SIBO) is objectively defined as

- an excess of bacteria $>10^5$ colony-forming units (CFU)/ml in duodenojejunal aspirate fluid ($>10^3$ CFU/ml if species identified are normally present in the colon—that is, gram negatives, enterococci, anaerobes), OR
- a glucose or lactose hydrogen breath test with either double peak or a peak of >20 parts per million (ppm) hydrogen above basal within 90 min of lactulose ingestion.

In the absence of objective data, symptoms such as abdominal pain, bloating, diarrhea, or flatulence that respond to a course of empiric antibiotic therapy can be used.

Background

SIBO is a poorly understood condition characterized by an excessive growth of bacteria within the small intestine resulting in a variety of GI and non-GI symptoms and complications. In children, SIBO is a common cause of nonspecific GI symptoms, such as chronic abdominal pain, abdominal distention, and diarrhea. Risk factors that may predispose children to SIBO include the use of acid-suppressive therapies and alterations in GI motility and anatomy.

Definition

Using the PubMed search term “small intestinal bacterial overgrowth,” articles were reviewed from January 1, 1970, until April 1, 2018. Only publications in English specifically focusing on the definition and classification of SIBO were selected. Any relevant publications retrieved from the references of the selected papers were included. To minimize duplicates, only articles with an “original” definition and classification were chosen. Of 249 articles reviewed, 101 defined SIBO either by breath test or by duodenal aspirate culture yield.

The most common definitions were

- >105 CFU/ml in jejunal aspirate fluid (51 articles);
- dual H₂ peak or >20 ppm increase in breath hydrogen (11 articles);
- >103 CFU/ml in jejunal aspirate fluid, provided the organisms normally colonize the colon (eight articles);
- >106 CFU/ml in jejunal aspirate fluid (eight articles); and
- >104 CFU/ml in jejunal aspirate fluid (three articles).

The remaining 18 articles used symptoms or simply increased bacterial content in the small bowel, without details.

SIBO is defined as an excess of bacteria >10⁵ CFU/ml in duodeno-jejunal aspirate fluid (>10³ CFU/ml if species identified are normally present in the colon—that is, gram negatives, enterococci, anaerobes) or as evidenced by a glucose or lactose hydrogen breath test with either double peak or a peak of >20 ppm hydrogen above basal within 90 min of lactulose ingestion.⁷⁸ This excess of bacteria and their metabolic by-products within the small bowel create a variety of symptoms in the GI tract such as abdominal pain, flatulence, and maldigestion.

In addition to simply having excessive bacteria in the small intestine, more-recent definitions include the presence of inappropriate microorganisms in specific regions of the small intestine. In healthy individuals, gram-positive aerobes are present in the proximal small bowel, whereas facultative anaerobes are found in the distal small bowel and anaerobes populate the colon.⁷⁹ In SIBO, the predominant organisms found in duodenojejunal aspirate cultures include *Bacteroides*, *Enterococcus*, and *Lactobacillus*.⁸⁰ In addition to the presence of bacterial overgrowth, there is also an imbalance in the usual ratios between these types of organisms.⁸¹

In the absence of these objective data, SIBO can be empirically defined as having at least three signs or symptoms that respond to a course of empiric antimicrobial therapy specifically targeted to manage

these signs and symptoms. This clinically diverse syndrome includes such GI symptoms as abdominal pain, abdominal bloating/distention, diarrhea, flatulence, and microscopic GI bleeding.⁸² If available, an intestinal biopsy showing villus atrophy, blunting, hyperplasia of crypts, and/or an increased number of lymphocytes in the lamina propria in the presence of symptoms or microscopic GI bleeding could be considered a tissue diagnosis of SIBO if symptoms/bleeding are responsive to antibiotic therapy.⁸³

The etiology of SIBO is multifactorial, and the most common causes include gastric achlorhydria, anatomic abnormalities within the small bowel resulting in stagnation, and small-bowel hypomotility. In healthy individuals, SIBO is prevented by the presence of the ileocecal valve, gastric acid, pancreatic enzymes, and motility in the small intestine.⁸⁴

There is a lack of a validated gold-standard test for diagnosing SIBO. Current methods of diagnosing SIBO by breath test involve administering an oral carbohydrate load and evaluating the types and number of organisms that produce metabolic by-products such as methane and hydrogen sulfide. The breath test has a sensitivity of 60%–90% and a specificity of 85% and continues to be the most commonly used diagnostic method, as it is the least invasive and most reproducible.⁸⁵ Depending upon the substrate used, false results are possible. For example, glucose, which is natively absorbed in the small intestine, when used as a diagnostic aid can result in more false-negative results, whereas lactulose, which is not absorbed, can lead to more false-positive findings. In 2017, a consensus document was developed that formalized cutoff values for exhaled hydrogen and methane.⁸⁶ These guidelines are not universally accepted, however. Many experts disagree, as data are conflicting and fail to take carbohydrate transit through the small intestine into consideration.⁸⁷ Use of breath tests in the pediatric population are not always feasible, especially in younger children, because of the need to fast prior to the test and the time necessary to perform the gas collection. In children with SBS, rapid transit time may further impact the interpretation of breath test results.⁸⁸

Aspiration and direct culturing of duodenojejunal contents can also be used to diagnose SIBO but is limited by its costs, invasiveness, and the risk of sample contamination by oropharyngeal bacteria during the intubation process. Moreover, overgrowth patterns may be inconsistent, and organisms may be missed with a single aspirate. In comparison with breath tests, culturing is less reproducible.^{89,90} In severe cases of SIBO, it may be accompanied by vitamin B₁₂ deficiencies due to competitive bacterial uptake or impaired absorption accompanied by an excess in folate production as a result of bacterial metabolism.⁹¹ Therefore, elevations in methylmalonic acid may also serve as a surrogate marker of SIBO.⁹²

Controversies and future directions

In addition to these classic tools used to diagnose bacterial overgrowth, other methods are being investigated to assess this condition. Piper et al have assessed stool samples from children with SBS for evidence of gut microbiota dysbiosis, using 16S ribosomal RNA sequencing and metagenomic shotgun sequencing.⁹³ They concluded

that patients with SBS complicated by overgrowth had a paucity of beneficial commensal anaerobes normally found in the gut microbiomes of healthy children.

Similarly, Mayeur et al studied 16 patients with SBS and showed a marked dysbiosis in fecal microbiota, with a predominance of the *Lactobacillus/Leuconostoc* group, whereas *Clostridium* and *Bacteroides* were underrepresented.⁹⁴ The presence of fecal lactate (56% of patients) was used to define a lactate-accumulator group, whereas an absence of fecal lactate (44% of patients) defined a non-lactate-accumulator group. The lactate-accumulator groups had lower serum bicarbonate levels and were at risk of D-lactate encephalopathy. Furthermore, all patients in the non-lactate-accumulator group and those accumulating preferentially L isoform in the lactate-accumulator group had never developed D-acidosis. The D/L fecal lactate ratio therefore may be a relevant index to predict patients with SIBO, particularly those at risk for D-lactic acidosis.

Another innovative approach that challenges the use of breath tests involves a capsule-based technology that provides real-time measurements of the major intraluminal gases (ie, hydrogen, carbon dioxide, oxygen, methane) as it moves through the gut.⁹⁵ Thus far, this research has been limited to healthy volunteers, but it has clearly demonstrated that an oral glucose load is incompletely absorbed and undergoes fermentation in the colon, resulting in a breath hydrogen peak (ie, the "early peak").

Taken together, these studies suggest bacterial function may be more important than the actual number of organisms; metabolomic studies may be necessary to understand SIBO and to refine its treatment. This is supported by a recent expert review on SIBO suggesting that the definition of SIBO should revert to simply stating that SIBO is "the presence of excessive numbers of bacteria in the small bowel, causing gastrointestinal symptoms."⁹⁶ The use of the absolute numbers for bacterial load within jejunal fluid or the use of breath tests for diagnosis is now undergoing reevaluation because of the variability in protocol design, the presence of confounding factors, and failure to arrive at a meaningful cutoff in the number and type of bacteria present in the small intestine.⁹⁷

Intestinal failure-associated liver disease

IFALD describes liver injury, as manifested by cholestasis, steatosis, and fibrosis, in patients with IF that is independent of, or in addition to, other potential etiologies. The development of IFALD is multifactorial, typically as a consequence of metabolic abnormalities in IF and the medical and surgical management strategies of IF themselves. It can be stabilized or reversed with appropriate early modification of management strategies and promotion of intestinal adaptation, or it can progress to hepatic dysfunction and end-stage liver disease.

Background

IFALD is characterized by a clinical spectrum ranging from mild steatosis, cholestasis, and fibrosis to liver failure with cirrhosis, portal

hypertension, and coagulopathy.⁹⁸ The etiology of IFALD is multifactorial, and risk factors include long-term PN, ILEs and especially plant sterols, recurrent septic episodes related to vascular access devices or small-bowel bacterial overgrowth, prematurity, and lack of EN.⁹⁸ The multifactorial pathogenesis of this liver impairment has led to the adoption of the term IFALD over "PN-associated liver disease/cholestasis (PNALD, PNAC)." Historically, IFALD was a common indication for liver and/or small-bowel transplantation and was one of the leading causes of death among this patient population.^{99–101} Avoidance and/or reversal of liver dysfunction remains the key challenge for clinicians working in this area and one of the goals of interdisciplinary treatment. Strategies used to prevent and treat IFALD include cycling of PN, sepsis prevention, use of alternative ILE, and small-bowel transplantation.^{102–106} The treatment of patients with IFALD has been fundamentally changed with the introduction of interdisciplinary intestinal rehabilitation programs (IRPs) and novel lipid management strategies such as lipid minimization or alternate lipid sources (eg, ω -3 ILEs).¹⁰⁷ Prevention of progression of IFALD provides the time necessary for intestinal adaptation to occur. No longer is there a "race against time" to wean from PN support before development of irreversible liver complications. Even patients with more extreme intestinal anatomy have an opportunity to adapt without transplantation and lifelong immunosuppression. For instance, in a long-term follow-up study on 100 patients with PIF with median PN duration of 1.2 years and median follow-up age of 6.6 years, 26% had derangements in liver biochemistry, 36% had METAVIR fibrosis stage 2 on liver biopsy, and none had progressive liver disease.¹⁰⁸ Among patients who demonstrate persistent evidence of IFALD, management in an IRP results in earlier detection, shorter time to transplant assessment, and improved outcomes among those patients who receive transplant.^{103,109} Two-thirds of patients with IF will develop IFALD, and traditionally, 25% would advance to end-stage liver disease. The long-term survival of PIF is 70%–90%, but the prevention of IFALD stands to improve the quality of life of children and their families. In spite of being the most important cause of morbidity and mortality, there has been no standardized definition of IFALD, and there is no agreed-upon clinical threshold by which to make the diagnosis.

Definition

A PubMed search was performed for the terms "intestinal failure associated liver disease" and "PN associated liver disease," producing 845 articles. Filters (review, human, English language, pediatrics) resulted in 48 articles. Clinically, IFALD is liver disease that is cholestatic and includes PN dependency and IF but omits other causes of liver disease. Therefore, there is not necessarily one pathognomonic finding biochemically, pathologically, or histologically. For adult patients, the European Society for Clinical Nutrition and Metabolism (ESPEN) proposed the definition of IFALD as "the presence of abnormal liver function tests and/or evidence of radiological and/or histological liver abnormalities in the absence of another primary parenchymal liver pathology" and also stated that abnormal liver histology is not mandatory to make the diagnosis.¹¹⁰

This raises the question: What is the appropriate biomarker for IFALD? The 2012 Gastroenterology Regulatory Endpoints and the Advancement of Therapeutics (GREAT): Parenteral Nutrition-Induced Liver Disease (PNALD) Workshop addressed this issue.¹¹¹ A biomarker should be objectively measured and should serve as an indicator of normal biological processes, disease processes, or pharmacologic responses to a therapeutic intervention. It should help establish a diagnosis, establish disease severity, or monitor progression. The desire is to link change in the biomarker to a clinically meaningful end point. As a clinical end point in an interventional study, it should be reproducible and responsive and change in the expected clinical direction. Although liver biopsy remains the gold standard for the evaluation of IFALD, the invasive nature of this test precludes its use as a frequent measure of disease progression or recovery. Conventional static biochemical liver tests (serum transaminases, bilirubin, alkaline phosphatase, and albumin levels and international normalized ratio [INR]) have been used to assess injury and function, but specificity and sensitivity varies considerably.^{112–114} For example, patients with IFALD with normal or mildly elevated bilirubin can often have cirrhosis as determined by liver biopsy.¹¹⁵

Roughly 50% of children receiving long-term PN will develop hepatic dysfunction, and the earliest clinical sign is a rise in conjugated bilirubin. It may start as early as within 2 weeks of starting PN and will rise during episodes of intercurrent sepsis. There may be a rise in alkaline phosphatase and aminotransferases within 4–6 weeks in about a third of the infants.¹¹¹ A persistent rise in alkaline phosphatase >1.5 times the normal upper limit has previously been considered "cholestatic," but this marker can be confounded by multiple tissue sources, and, in older patients, it equally correlates with the degree of steatosis.¹¹⁶ Biochemistry, however, can remain remarkably static or normal until end-stage disease emerges unexpectedly, often marked only by a rapid elevation of serum bilirubin. One reason for the late presentation of end-stage IFALD is that with reduced portal inflow to the liver (as seen in SBS), the portal hypertension signs of esophageal varices, encephalopathy, and ascites do not occur. Regardless, for practical reasons, serum conjugated bilirubin has emerged as the most common test to diagnose IFALD. It has the benefit of being inexpensive, readily available, easy to measure, and widely recognized. Serum bile acids are actually an earlier indicator of cholestasis, but this is not readily available in many centers and is prone to wide population variability due to polymorphisms in bile acid transport.¹¹¹ The elevation of transaminases and/or gamma-glutamyltransferase (>1.5 upper limit of normal) has been proposed in some studies.¹¹⁷ The aspartate aminotransferase/platelet ratio has been utilized as a predictor of need for transplantation, but not as a definition of IFALD.

Composite scoring systems have shown utility in assessing chronic liver disease. The "model for end-stage liver disease" (MELD) was developed to predict survival in adult patients with chronic liver disease using serum bilirubin, creatinine, and INR.¹¹⁸ The pediatric end-stage liver disease (PELD) score was designed as a tool for predicting mortality in children with chronic disease awaiting liver transplantation.¹¹⁹ PELD score employs criteria from the MELD to highlight the unique growth and development needs of children. These

include bilirubin, INR, and serum albumin level (to measure the liver's synthetic function); growth failure; and age (>1 or <1 year). Composite scoring systems are potentially better than individual biochemical tests, as they consider several variables that reflect hepatic reserve and function. However, although the PELD score is specific to the pediatric population, it is not validated for IFALD. Importantly, several of the components of PELD (weight, height, INR, and serum albumin level) can be impacted by conditions other than IFALD. For example, prematurity, gestational age, malabsorption, intestinal losses, intestinal length, and malnutrition may contribute to abnormalities of growth, vitamin K absorption, and hypoalbuminemia independent of liver function.

There have been numerous attempts to monitor liver disease without the need for biopsy, including the use of ultrasound elastography ("Fibroscan") and stable isotopic assessment of liver function.^{120–123} Although promising, these have not yet shown a definitive ability to identify progressive liver disease in this setting and cannot yet be relied on clinically. At the current time, liver biopsy remains the best modality for identifying and monitoring the progression of the condition, despite its obvious limitations. Children with cirrhosis can remain stable for long periods.³³ For practical reasons, liver biopsy can be performed most conveniently at the time of other abdominal procedures such as autologous reconstruction to obtain current information of liver histology. When available, histology is characterized by cholestasis, features of biliary obstruction (portal inflammation, edema, ductular proliferation), and microvesicular and macrovesicular zone 1 steatosis. In older patients, steatohepatitis can also be seen. In advanced disease, fibrosis has a biliary pattern, beginning with portal expansion and progressing to periportal fibrosis, ultimately ending in bridging fibrosis.¹²⁴ Steatosis is often also associated with this later fibrotic stage.¹²⁵

Despite the lack of a formal definition of IFALD, review of the literature has revealed that many centers have gradually adopted a conjugated bilirubin of 2 mg/dl (34 μ mol/L) as the definition of IFALD.^{111,126} This threshold has, however, not been universal and is not supported by strong evidence but rather an adoption over time by many researchers. There has been some variation with this, as some centers have utilized a serum conjugated bilirubin level >3 mg/dl (50 μ mol/L) for >2 weeks and not associated with a concurrent sepsis episode as a definition of IFALD.¹²⁷ The European Society for Paediatric Gastroenterology, Hepatology and Nutrition (ESPGHAN) position paper on IF and transplantation noted conjugated bilirubin level of >20 μ mol/L (1.2 mg/dl) for 2–4 weeks as a "red flag" for early IFALD.¹²⁸ In addition, many studies have also used 2 mg/dl as a threshold to initiate alternative lipid management strategies. For research purposes, it is important to agree on a threshold value to allow reporting of this important clinical outcome. As a result, the panel has agreed that a conjugated bilirubin of 2 mg/dl (34 μ mol/L) sustained for 2 weeks, without concurrent sepsis, after PN exposure of a minimum of 2 weeks is a reasonable definition of IFALD. Other causes of liver disease must be ruled out, including infectious hepatitis, metabolic diseases, cystic fibrosis, and anatomical abnormalities. Further controversy exists, however, about whether this threshold represents mild or severe disease. In 2005, Nasr et al demonstrated that in 292 surgical neonates receiving PN, 104 (36%) reached a conjugated bilirubin level of 34 μ mol/L (2 mg/dl) in the era immediately prior

to the introduction of alternate lipid emulsions.¹²⁹ Eighty-six patients (83%) weaned off PN, 31 patients (30%) developed more-advanced IFALD with a conjugated bilirubin level of 100 $\mu\text{mol/L}$ (6 mg/dl), and nine patients (9%) either died or were transplanted.¹²⁹ Therefore, a conjugated bilirubin level of 2 mg/dl (34 $\mu\text{mol/L}$) reasonably represents early IFALD. Christensen et al demonstrated in neonates dependent on PN that the incidence of IFALD, defined by a conjugated bilirubin level >2 mg/dl, was 14% at days 14–28, 44% at days 29–56, 72% at days 57–100, and >85% after >100 days.¹³⁰

In addition, we propose advanced IFALD could be defined as a conjugated bilirubin level >6 mg/dl (100 $\mu\text{mol/L}$) sustained for 2 weeks, without concurrent sepsis, after PN exposure for a minimum of 2 weeks, as this value has been used in previous literature.^{127,128,131–134} The difficulty is incorporating histologic findings into such a diagnosis. Fitzgibbons et al correlated liver biochemistry with liver biopsy in patients with PIF and demonstrated that reversal of PN-associated cholestasis can be observed despite persistent fibrosis or even cirrhosis.¹¹⁵ Eighty-nine percent of liver biopsies (74 of 83) demonstrated some degree of fibrosis (fibrosis scale 1–3). More than half (55%) of these cases were obtained from patients without biochemical evidence of cholestasis. Almost 10% ($n=8$) of all biopsy specimens had evidence of cirrhosis.¹¹⁵ Liver failure has previously been defined as the presence of any two of the following: sustained serum conjugated bilirubin level >200 $\mu\text{mol/L}$ for at least 2 weeks (not associated with sepsis), an INR >1.5, serum albumin level <20 g/L, platelets <100,000, evidence of portal hypertension (splenomegaly, varices), or bridging fibrosis on liver biopsy.¹²⁷ The ESPGHAN position paper stated liver failure included patients exhibiting portal hypertension, varices, splenomegaly, and coagulopathy often with elevated bilirubin levels and thrombocytopenia.¹²⁸ The refinement of the definition of advanced IFALD or liver failure to include the other parameters listed above will need to be the focus of future work.

Controversies and future directions

One of the major challenges facing clinicians managing patients with PIF is an emerging liver disease in which cholestasis has resolved but patients have underlying progressive fibrosis with continued dependence on PN support. The development of noncholestatic liver disease with portal hypertension due to progressive fibrosis poses significant management challenges. This is in contrast to our historical experience in which patients developed progressive cholestasis that could be followed biochemically. This new phenotype may be the result of improved bile flow with the use of alternative lipid strategies in the setting of a systemic proinflammatory milieu. The diagnosis of IFALD requires the synthesis of clinical, biochemical, radiological, and, where appropriate, histological information. Noninvasive monitoring permitting early detection and treatment options that avoid progressive liver injury is the goal for patients with PIF on chronic PN. In addition, intestinal transplantation should be considered early in the face of progressive liver disease. Noninvasive assessment, if feasible and accurate, provides the possibility to frequently assess IFALD and give more

insight into its progression, enable evaluation of the effects of interventions, and enable the identification of individuals at risk for cirrhosis and its complications.

Emerging imaging technologies such as transient elastography (TE) may prove useful as a noninvasive measurement of liver fibrosis. TE measurements are based on acquisition of pulse-echo ultrasound signals to measure liver stiffness.¹³⁵ Liver stiffness is used as an indicator of liver fibrosis, but in addition to fibrosis, liver stiffness is also determined by extrahepatic cholestasis and liver congestion.¹³⁵ TE compared with liver biopsy has been used to determine stages of fibrosis in both adults and children receiving home PN.^{121–123} TE failed to accurately assess the degree of hepatic fibrosis, but TE values did correlate with serum bilirubin levels and degree of cholestasis. Further evaluation of this tool in children may provide another method of evaluating progressive liver disease in PIF, as biochemical markers alone cannot be relied upon in this setting.

Quantitative methionine breath test to assess hepatic function is promising, but early in development. In this breath test, the patient is administered IV [$1\text{-}^{13}\text{C}$]-Met, in which one of the carbon atoms in methionine has been replaced by a stable ^{13}C isotope. The ^{13}C stable isotope is metabolized and then incorporated into CO_2 and exhaled.¹²⁰ This allows one to follow methionine oxidative capacity in the mitochondria of hepatocytes. Duro et al presented 27 patients (median age 5.3 months) who underwent a total of 34 [^{13}C]-Met breath tests. In 14 patients who had documented liver biopsies, the [^{13}C]-Met breath test was able to distinguish patients with and without cirrhosis. Serial [^{13}C]-Met breath tests were performed in five patients and correlated with change in liver function over time as determined by the PELD score. The methionine breath test offers the potential for a safe, noninvasive means of evaluating IFALD in children, and it may also help quantify progression or improvement over time.¹²⁰

Although liver biopsy continues to be the current gold standard in the diagnosis of cirrhosis, the accuracy of liver biopsy is also limited, owing to the lack of specific histological grading and staging of IFALD, which can result in significant intraobserver and interobserver variability.¹³⁶ Validation of a histological scoring system that is specific for IFALD is needed so that when liver biopsies are performed, the results are consistently reported and, therefore, more comparable between research studies.

Intestinal rehabilitation program

An IRP is an interdisciplinary, collaborative patient care paradigm that serves to coordinate care for children with IF through comprehensive management of their specialized nutrition and corollary needs, attention to and support for associated chronic comorbidities, and evaluation and treatment of acute complications.

Background

The care of complex patients, in general, has recently moved toward interdisciplinary models in which patients are managed by a group

of interested and expert providers within a single organizational construct. Within PIF, these programs have been found to have a significant impact, even demonstrating improvement in survival.^{103,137,138} It has been the observation of individuals and organizations involved in the care of patients with PIF that IRPs are now becoming increasingly established around the country, but with varying resources and capabilities and goals. With this variability, the ASPEN PIF Section felt that a standardization of the definition of an IRP and a discussion about the requirements, goals, and future development of IRPs were needed in the context of this document.

Definition

PubMed was queried for the search terms “PIF” and “pediatric short bowel syndrome,” using the standard English language and human participant filters, including 1970 to March 2018. Abstracts were reviewed, duplicates were eliminated, and then items felt to be topically pertinent were reviewed in full, with a bias toward inclusion. Of 270 manuscripts reviewed, ones without an explicit definition of an IF center or program were eliminated, and references for those included were evaluated for additional inclusion. Ultimately, 36 manuscripts were reviewed in total, with many redundant or nonoriginal definitions, with all of the original manuscripts referenced below. In addition to this sparse reference material available, the definitions working group as well as the members of the PIF Section were solicited for expert opinion on the matter.

Although many synonyms can be and have been used, we define a pediatric IRP as an interdisciplinary, collaborative patient care paradigm that serves to coordinate care for children with IF through comprehensive management of their specialized nutrition and corollary needs, attention to and support for associated chronic comorbidities, and evaluation and treatment of acute complications.

IRPs accomplish these stated goals through several critical components:

1. Interdisciplinary evaluation and management by key personnel and integral available consultants (see below)
2. Integration of long-term specialized nutrition services, including inpatient and outpatient PN, tailored EN formulas and administration, and medical therapy to maximize intestinal adaptation
3. The ability to serve as resources for referral of complex PIF patients from institutions local to the patient that do not have the necessary expertise or capabilities
4. The creation of quality improvement-driven protocols for management of complex PIF, such as advancement of EN, provision and coordination of inpatient and outpatient PN, promotion of adaptation through medical therapy, appropriate application of surgical intervention for anatomic disorders as well as autologous intestinal reconstructive surgery, appropriate application of advanced GI assessment through radiologic and endoscopic means, serial monitoring of diagnostic markers for long-term comorbidities, and others

Although it is recognized that IRPs will have different individual arrangements based on their local environment, the key to all IRPs is the ability to have streamlined communication among care providers that is ideally carried out through a single interdisciplinary inpatient and outpatient structure and allows for “one-stop” shopping. IRPs accomplish these stated goals with the following integral key personnel. Although a single provider may fulfill multiple roles listed below, we have laid out the roles individually to highlight the functions that are considered part of an IRP, both in the “essential” and “key” personnel sections^{33,139}:

1. Personnel/functions considered essential or mandatory to an IRP
 - a. Pediatric gastroenterologist
 - b. Pediatric surgeon
 - c. PN physician (may be same as another physician)
 - d. Dietitian
 - e. Nurse coordinator
 - f. Program administrator
 - g. Pharmacist
 - h. Social worker
2. Other key personnel who are critical and preferable but may not be a standing member of the IRP (ie, available as consultant) or may not be available at all IRPs
 - a. Physicians
 - i. Transplant (GI, hepatology, surgery)
 - ii. Subspecialists
 1. Radiology/interventional radiology
 2. Nephrology
 3. Endocrinology
 4. Pathology
 5. Hepatology
 6. Hematology
 7. Neonatology
 8. Infectious disease
 - iii. Trainees
 - b. Nursing (individual IRPs will have their own structures)
 - i. Nurse practitioner(s): often one is the center coordinator
 - ii. Nurses
 1. General nursing care: phlebotomy, dressing care, vaccines/preventive care
 2. Stoma/wound/enteral tube specialists: enteral tube care/changes, stoma care/appliance management
 - iii. Care coordination
 - iv. Separate home PN nurses
 - c. Administrative
 - i. Database administrator
 - d. Speech and language pathology/occupational therapy/feeding therapy
 - e. Research team
 - i. Research fellows
 - ii. Research assistant(s) or coordinator(s)
 - f. Clinical psychologist
 - g. Child life specialist

Management at an IRP focuses initially on early care at the time of intestinal injury through consultation, direct oversight of care, or a less structured interaction with primary providers, as dictated by the local environment. This management guides nutrition care with a long-term vision, encouraging early EN, monitoring for micronutrient and vitamin deficiencies, and appropriate parenteral support strategies such as hepatoprotection, cycling of PN when able, choice of ILE, and others. This early management also stresses surgical guidance through consultation or direct care, focusing on appropriate diagnostic studies and differential diagnoses, bowel preservation, and decision making around ostomy creation/reversal and vascular access device care.

This early care can transition through introduction to the interdisciplinary model to long-term care, either as the primary center of care or as a referral location for outside institutions. This longitudinal care enhances nutrition management with the goal of medical optimization, weaning of PN, advancement of enteral tolerance, and monitoring for nutrient deficiencies. Long-term surgical care focuses on bowel continuity and the assessment for potential complications, including stricture, stoma-related difficulties, and management of vascular access devices and EADs. Surgical intervention for autologous intestinal reconstruction through tapering and lengthening operations, such as serial transverse enteroplasty, can also play a pivotal role in the advancement toward enteral autonomy.^{140,141} Finally, patients with PIF are at risk for multiple potential comorbidities, and the IRP is tasked with appropriate screening, monitoring, and management of these issues, both in the clinic and with the assistance of subspecialty consultants.^{33,139}

Controversies and future directions

The PIF community continues to grow, and IRPs continue to form. The future (and present) goals in regards to IRPs center around the appropriate development and success of IRPs. Appropriate criteria to identify patients that should be referred to an IRP are critical to allow for early management, as discussed above, which can transition to long-term care and can potentially prevent or mitigate long-term complications. Multiple authors, including the members of the working group, advocate for any child meeting the criteria for PIF (as indicated in the section above) being referred to an IRP.¹⁴² Ideally, predicting which patients will go on to qualify for a diagnosis of PIF would allow for even earlier referral. This is especially true for the prevention and treatment of IFALD, for which expertise and interdisciplinary care can prevent progression, allow reversal, and result in improved survival.^{137,142} In many cases, IRPs can function in parallel with providers local to the patient through consultation or collaborative management to optimize care and minimize burden. The local provider is, in fact, key to the success of IRPs, which may not be in close geographic proximity to the patient, especially during acute presentations for dehydration, fever, central line-associated bloodstream infection, etc. This allows for close communication among the patient, IRP, and local provider, which can allow for rapid assessment when necessary, appropriate diagnostic

testing and care, and a discussion about need for in-person evaluation at the IRP vs management by the local care provider guided by the IRP.

Recent focus on the successful development of IRPs merits discussion here and in future studies. The goal of every new IRP should be to serve as a state-of-the-art referral center aimed at minimizing complications and maximizing progress toward the goal of weaning from specialized nutrition needs. This can be measured by improved survival, decreased morbidity, and improved quality of life.^{6,143,144} The components needed to achieve this goal, as highlighted above, can be considered individually for each local environment, with some clearly being essential and some becoming more needed as a center grows or becomes more resourced. Central support at the institutional leadership level is paramount, and early success must be demonstrated to foster this support and justify it. Frequent check-ins with institutional administration, as well as demonstration of success through patient volume and academic productivity (publication, research, registry development), are helpful in this regard. To that end, IRPs internationally have contributed to most of the information in this manuscript and more, such as demonstration of improved survival,¹³⁷ use of ethanol locks for the prevention of central line-associated bloodstream infections,^{145,146} and alternate lipid management strategies.¹⁴⁷⁻¹⁴⁹

CONCLUSION

It is the hope of ASPEN and the PIF Section that this document will lead to standardization of the terms and definitions contained herein and to more-consistent reporting of clinical outcomes and research results to allow for consistent and reliable discourse in the improvement of management of complex patients with PIF. It is expected that this will be a reiterative process, with regular updating, revision, and expansion required to incorporate new information and include more key terms in the library of standardized definitions within PIF.

CONFLICT OF INTEREST

Kathleen Gura is a consultant and advisory board member for Baxter.

FUNDING INFORMATION

KathleenGura has a patent and licensing agreement with and has received grant/research/clinical trial support from Fresenius Kabi and grant/research/clinical trial support from Otsuka and NorthSea Therapeutics.

ORCID

Biren P. Modi MD, MPH  <https://orcid.org/0000-0003-2736-2186>

David P. Galloway MD  <https://orcid.org/0000-0001-8811-3111>

Kathleen Gura PharmD  <https://orcid.org/0000-0002-0431-896X>

Anita Nucci PhD, RD  <https://orcid.org/0000-0001-8823-7384>

REFERENCES

1. Diamond IR, Grant RC, Feldman BM, et al. Defining consensus: a systematic review recommends methodologic criteria for reporting of Delphi studies. *J Clin Epidemiol*. 2014;67(4):401-409.

2. Pironi L, Arends J, Baxter J, et al. ESPEN endorsed recommendations. Definition and classification of intestinal failure in adults. *Clin Nutr*. 2015;34(2):171-180.
3. Barclay AR, Beattie LM, Weaver LT, Wilson DC. Systematic review: medical and nutritional interventions for the management of intestinal failure and its resultant complications in children. *Aliment Pharmacol Ther*. 2011;33(2):175-184.
4. Chan CF, Wu TC. Recent advances in the management of pediatric intestinal failure. *Pediatr Neonatol*. 2014;55(6):426-430.
5. Ching YA, Gura K, Modi B, Jaksic T. Pediatric intestinal failure: nutrition, pharmacologic, and surgical approaches. *Nutri Clin Pract*. 2007;22(6):653-663.
6. Duggan CP, Jaksic T. Pediatric intestinal failure. *N Engl J Med*. 2017;377(7):666-675.
7. Gosselin KB, Duggan C. Enteral nutrition in the management of pediatric intestinal failure. *J Pediatr*. 2014;165(6):1085-1090.
8. Goulet O, Ruemmele F, Lacaille F, Colomb V. Irreversible intestinal failure. *J Pediatr Gastroenterol Nutr*. 2004;38(3):250-269.
9. Gupte GL, Beath SV, Kelly DA, Millar AJ, Booth IW. Current issues in the management of intestinal failure. *Arch Dis Child*. 2006;91(3):259-264.
10. Kaufman SS, Matsumoto CS. Management of pediatric intestinal failure. *Minerva Pediatr*. 2015;67(4):321-340.
11. Wong T, Gupte G. Intestinal failure in children. *Indian J Pediatr*. 2016;83(12-13):1436-1443.
12. Duro D, Kamin D, Duggan C. Overview of pediatric short bowel syndrome. *J Pediatr Gastroenterol Nutr*. 2008;47(Suppl 1):S33-36.
13. Squires RH, Duggan C, Teitelbaum DH, et al. Natural history of pediatric intestinal failure: initial report from the pediatric intestinal failure consortium. *J Pediatr*. 2012;161(4):723-728.e2.
14. Batra A, Keys SC, Johnson MJ, Wheeler RA, Beattie RM. Epidemiology, management and outcome of ultrashort bowel syndrome in infancy. *Arch Dis Child Fetal Neonatal Ed*. 2017;102(6):F551-F556.
15. Bhatia J, Gates A, Parish A. Medical management of short gut syndrome. *J Perinatol*. 2010;30 Suppl:S2-S5.
16. Booth IW, Lander AD. Short bowel syndrome. *Baillieres Clin Gastroenterol*. 1998;12(4):739-773.
17. Coletta R, Khalil BA, Morabito A. Short bowel syndrome in children: surgical and medical perspectives. *Semin Pediatr Surg*. 2014;23(5):291-297.
18. Fernandes MA, Usatin D, Allen IE, Rhee S, Vu L. Improved enteral tolerance following step procedure: systematic literature review and meta-analysis. *Pediatr Surg Int*. 2016;32(10):921-926.
19. Goulet O, Olieman J, Ksiazky J, et al. Neonatal short bowel syndrome as a model of intestinal failure: physiological background for enteral feeding. *Clin Nutr*. 2013;32(2):162-171.
20. Goulet O, Sauvat F. Short bowel syndrome and intestinal transplantation in children. *Curr Opin Clin Nutr Metab Care*. 2006;9(3):304-313.
21. Gutierrez IM, Kang KH, Jaksic T. Neonatal short bowel syndrome. *Semin Fetal Neonatal Med*. 2011;16(3):157-163.
22. Schwartz MZ. Novel therapies for the management of short bowel syndrome in children. *Pediatr Surg Int*. 2013;29(10):967-974.
23. Serrano MS, Schmidt-Sommerfeld E. Nutrition support of infants with short bowel syndrome. *Nutrition*. 2002;18(11-12):966-970.
24. Stringer MD, Puntis JW. Short bowel syndrome. *Arch Dis Child*. 1995;73(2):170-173.
25. Sukhotnik I, Siplovich L, Shiloni E, Mor-Vaknin N, Harmon CM, Coran AG. Intestinal adaptation in short-bowel syndrome in infants and children: a collective review. *Pediatr Surg Int*. 2002;18(4):258-263.
26. Vanderhoof JA, Langnas AN. Short-bowel syndrome in children and adults. *Gastroenterology*. 1997;113(5):1767-1778.
27. Vanderhoof JA, Young RJ, Thompson JS. New and emerging therapies for short bowel syndrome in children. *Paediatr Drugs*. 2003;5(8):525-531.
28. Batra A, Beattie RM. Management of short bowel syndrome in infancy. *Early Hum Dev*. 2013;89(11):899-904.
29. Burghardt KM, Wales PW, de Silva N, et al. Pediatric intestinal transplant listing criteria - a call for a change in the new era of intestinal failure outcomes. *Am J Transplant*. 2015;15(6):1674-1681.
30. Roy CC, Groleau V, Bouthillier L, Pineault M, Thibault M, Marchand V. Short bowel syndrome in infants: the critical role of luminal nutrients in a management program. *Appl Physiol Nutr Metab*. 2014;39(7):745-753.
31. Wales PW, Nasr A, de Silva N, Yamada J. Human growth hormone and glutamine for patients with short bowel syndrome. *Cochrane Database Syst Rev*. 2010(6):CD006321.
32. Weih S, Kessler M, Fonouni H, et al. Current practice and future perspectives in the treatment of short bowel syndrome in children—a systematic review. *Langenbecks Arch Surg*. 2012;397(7):1043-1051.
33. Merritt RJ, Cohran V, Raphael BP, et al. Intestinal rehabilitation programs in the management of pediatric intestinal failure and short bowel syndrome. *J Pediatr Gastroenterol Nutr*. 2017;65(5):588-596.
34. Cohran VC, Prozialeck JD, Cole CR. Redefining short bowel syndrome in the 21st century. *Pediatr Res*. 2017;81(4):540-549.
35. Dore M, Junco PT, Moreno AA, et al. Ultrashort bowel syndrome outcome in children treated in a multidisciplinary intestinal rehabilitation unit. *Eur J Pediatr Surg*. 2017;27(1):116-120.
36. Infantino BJ, Mercer DF, Hobson BD, et al. Successful rehabilitation in pediatric ultrashort small bowel syndrome. *J Pediatr*. 2013;163(5):1361-1366.
37. Sanchez SE, Javid PJ, Healey PJ, Reyes J, Horslen SP. Ultrashort bowel syndrome in children. *J Pediatr Gastroenterol Nutr*. 2013;56(1):36-39.
38. Hong CR, Han SM, Staffa SJ, Carey AN, Modi BP, Jaksic T. Long-term outcomes of ultrashort bowel syndrome due to malrotation with midgut volvulus managed at an interdisciplinary pediatric intestinal rehabilitation center. *J Pediatr Surg*. 2019;54(5):964-967.
39. Struijs MC, Diamond IR, de Silva N, Wales PW. Establishing norms for intestinal length in children. *J Pediatr Surg*. 2009;44(5):933-938.
40. Dudrick SJ. History of parenteral nutrition. *J Am Coll Nutr*. 2009;28(3):243-251.
41. Latta T. Letter from Dr. Latta to the Secretary of the Central Board of Health, London, affording a view of the rationale and results of his practice in the treatment of cholera by aqueous and saline injections. 1832. *Int J Epidemiol*. 2013;42(2):387-390.
42. Vanek VW, Borum P, Buchman A, et al. A.S.P.E.N. position paper: recommendations for changes in commercially available parenteral multivitamin and multi-trace element products. *Nutri Clin Pract*. 2012;27(4):440-491.
43. Namjoshi SS, Muradian S, Bechtold H, et al. Nutrition deficiencies in children with intestinal failure receiving chronic parenteral nutrition. *JPEN J Parenter Enteral Nutr*. 2018;42(2):427-435.
44. Cederholm T, Barazzoni R, Austin P, et al. ESPEN guidelines on definitions and terminology of clinical nutrition. *Clin Nutr*. 2017;36(1):49-64.
45. Druml C, Ballmer PE, Druml W, et al. ESPEN guideline on ethical aspects of artificial nutrition and hydration. *Clin Nutr*. 2016;35(3):545-556.
46. Dibb M, Teubner A, Theis V, Shaffer J, Lal S. Review article: the management of long-term parenteral nutrition. *Aliment Pharmacol Ther*. 2013;37(6):587-603.
47. Iyer KR, Kunecki M, Boullata JL, et al. Independence from parenteral nutrition and intravenous fluid support during treatment with teduglutide among patients with intestinal failure associated with short bowel syndrome. *JPEN J Parenter Enteral Nutr*. 2017;41(6):946-951.
48. Pironi L. Definitions of intestinal failure and the short bowel syndrome. *Best Pract Res Clin Gastroenterol*. 2016;30(2):173-185.
49. Amin SC, Pappas C, Iyengar H, Maheshwari A. Short bowel syndrome in the NICU. *Clin Perinatol*. 2013;40(1):53-68.

50. Nucci AM, Ellsworth K, Michalski A, Nagel E, Wessel J; ASPEN Pediatric Intestinal Failure Section. Survey of nutrition management practices in centers for pediatric intestinal rehabilitation. *Nutr Clin Pract*. 2018;33(4):528-538.
51. Avitzur Y, Courtney-Martin G. Enteral approaches in malabsorption. *Best Pract Res Clin Gastroenterol*. 2016;30(2):295-307.
52. Savoie KB, Bachier-Rodriguez M, Jones TL, et al. Standardization of feeding advancement after neonatal gastrointestinal surgery: does it improve outcomes?. *Nutr Clin Pract*. 2016;31(6):810-818.
53. Shores DR, Alaish SM, Aucott SW, et al. Postoperative enteral nutrition guidelines reduce the risk of intestinal failure-associated liver disease in surgical infants. *J Pediatr*. 2018;195:140-147.e1.
54. Andorsky DJ, Lund DP, Lillehei CW, et al. Nutritional and other post-operative management of neonates with short bowel syndrome correlates with clinical outcomes. *J Pediatr*. 2001;139(1):27-33.
55. Cole CR, Kocoshis SA. Nutrition management of infants with surgical short bowel syndrome and intestinal failure. *Nutr Clin Pract*. 2013;28(4):421-428.
56. Harvie ML, Norris MAT, Sevilla WMA. Soluble fiber use in pediatric short bowel syndrome: a survey on prevailing practices. *Nutr Clin Pract*. 2018;33(4):539-544.
57. Yang CF, Duro D, Zurakowski D, Lee M, Jaksic T, Duggan C. High prevalence of multiple micronutrient deficiencies in children with intestinal failure: a longitudinal study. *J Pediatr*. 2011;159(1):39-44 e31.
58. Ubesie AC, Kocoshis SA, Mezoff AG, Henderson CJ, Helmrath MA, Cole CR. Multiple micronutrient deficiencies among patients with intestinal failure during and after transition to enteral nutrition. *J Pediatr*. 2013;163(6):1692-1696.
59. DiMaggio DM, Cox A, Porto AF. Updates in infant nutrition. *Pediatr Rev*. 2017;38(10):449-462.
60. Boullata JI, Carrera AL, Harvey L, et al; ASPEN Safe Practices for Enteral Nutrition Therapy Task Force, American Society for Parenteral and Enteral Nutrition. ASPEN safe practices for enteral nutrition therapy [Formula: see text]. *JPEN J Parenter Enteral Nutr*. 2017;41(1):15-103.
61. Geertsma MA, Hyams JS, Pelletier JM, Reiter S. Feeding resistance after parenteral hyperalimentation. *Am J Dis Child*. 1985;139(3):255-256.
62. Schwartz MZ, Maeda K. Short bowel syndrome in infants and children. *Pediatr Clin North Am*. 1985;32(5):1265-1279.
63. Vanderhoof JA. Short bowel syndrome in children and small intestinal transplantation. *Pediatr Clin North Am*. 1996;43(2):533-550.
64. Demehri FR, Stephens L, Herrman E, et al. Enteral autonomy in pediatric short bowel syndrome: predictive factors one year after diagnosis. *J Pediatr Surg*. 2015;50(1):131-135.
65. Ives GC, Demehri FR, Sanchez R, Barrett M, Gadepalli S, Teitelbaum DH. Small bowel diameter in short bowel syndrome as a predictive factor for achieving enteral autonomy. *J Pediatr*. 2016;178:275-277.e1.
66. Jeppesen PB, Mortensen PB. Colonic digestion and absorption of energy from carbohydrates and medium-chain fat in small bowel failure. *JPEN J Parenter Enteral Nutr*. 1999;23(5 Suppl):S101-S105.
67. Khan FA, Squires RH, Litman HJ, et al. Predictors of enteral autonomy in children with intestinal failure: a multicenter cohort study. *J Pediatr*. 2015;167(1):29-34 e21.
68. Quiros-Tejiera RE, Ament ME, Reyen L, et al. Long-term parenteral nutritional support and intestinal adaptation in children with short bowel syndrome: a 25-year experience. *J Pediatr*. 2004;145(2):157-163.
69. Rudolph JA, Squires R. Current concepts in the medical management of pediatric intestinal failure. *Curr Opin Organ Transplant*. 2010;15(3):324-329.
70. Sparks EA, Khan FA, Fisher JG, et al. Necrotizing enterocolitis is associated with earlier achievement of enteral autonomy in children with short bowel syndrome. *J Pediatr Surg*. 2016;51(1):92-95.
71. Cooper A, Floyd TF, Ross AJ III, Bishop HC, Templeton JM Jr, Ziegler MM. Morbidity and mortality of short-bowel syndrome acquired in infancy: an update. *J Pediatr Surg*. 1984;19(6):711-718.
72. Goulet O, Baglin-Gobet S, Talbotec C, et al. Outcome and long-term growth after extensive small bowel resection in the neonatal period: a survey of 87 children. *Eur J Pediatr Surg*. 2005;15(2):95-101.
73. Simon NP. Follow-up for infants with necrotizing enterocolitis. *Clin Perinatol*. 1994;21(2):411-424.
74. McLaughlin CM, Channabasappa N, Pace J, Nguyen H, Piper HG. Growth trajectory in children with short bowel syndrome during the first 2 years of life. *J Pediatr Gastroenterol Nutr*. 2018;66(3):484-488.
75. Mehta NM, Corkins MR, Lyman B, et al. Defining pediatric malnutrition: a paradigm shift toward etiology-related definitions. *JPEN J Parenter Enteral Nutr*. 2013;37(4):460-481.
76. Frongia G, Kessler M, Weih S, Nickkholgh A, Mehrabi A, Holland-Cunz S. Comparison of LILT and STEP procedures in children with short bowel syndrome—a systematic review of the literature. *J Pediatr Surg*. 2013;48(8):1794-1805.
77. Ramos-Gonzalez G, Kim HB. Autologous intestinal reconstruction surgery. *Immun Pediatr Surg*. 2018;27(4):261-266.
78. Gutierrez IM, Kang KH, Calvert CE, et al. Risk factors for small bowel bacterial overgrowth and diagnostic yield of duodenal aspirates in children with intestinal failure: a retrospective review. *J Pediatr Surg*. 2012;47(6):1150-1154.
79. Tropini C, Earle KA, Huang KC, Sonnenburg JL. The gut microbiome: connecting spatial organization to function. *Cell Host Microbe*. 2017;21(4):433-442.
80. Bouhnik Y, Alain S, Attar A, et al. Bacterial populations contaminating the upper gut in patients with small intestinal bacterial overgrowth syndrome. *Am J Gastroenterol*. 1999;94(5):1327-1331.
81. Dukowicz AC, Lacy BE, Levine GM. Small intestinal bacterial overgrowth: a comprehensive review. *Gastroenterol Hepatol (N Y)*. 2007;3(2):112-122.
82. Grace E, Shaw C, Whelan K, Andreyev HJ. Review article: small intestinal bacterial overgrowth—prevalence, clinical features, current and developing diagnostic tests, and treatment. *Aliment Pharmacol Ther*. 2013;38(7):674-688.
83. Lappinga PJ, Abraham SC, Murray JA, Vetter EA, Patel R, Wu TT. Small intestinal bacterial overgrowth: histopathologic features and clinical correlates in an underrecognized entity. *Arch Pathol Lab Med*. 2010;134(2):264-270.
84. Bures J, Cyrany J, Kohoutova D, et al. Small intestinal bacterial overgrowth syndrome. *World J Gastroenterol*. 2010;16(24):2978-2990.
85. Riordan SM, McIver CJ, Walker BM, Duncombe VM, Bolin TD, Thomas MC. The lactulose breath hydrogen test and small intestinal bacterial overgrowth. *Am J Gastroenterol*. 1996;91(9):1795-1803.
86. Rezaie A, Buresi M, Lembo A, et al. Hydrogen and methane-based breath testing in gastrointestinal disorders: the North American consensus. *Am J Gastroenterol*. 2017;112(5):775-784.
87. Paterson W, Camilleri M, Simren M, Boeckstaens G, Vanner SJ. Breath testing consensus guidelines for SIBO: RES IPSA LOC-QUITOR. *Am J Gastroenterol*. 2017;112(12):1888-1889.
88. Avelar Rodriguez D, Ryan PM, Toro Monjaraz EM, Ramirez Mayans JA, Quigley EM. Small intestinal bacterial overgrowth in children: a state-of-the-art review. *Front Pediatr*. 2019;7:363.
89. Braun-Moscovici Y, Braun M, Khanna D, Balbir-Gurman A, Furst DE. What tests should you use to assess small intestinal bacterial overgrowth in systemic sclerosis?. *Clin Exp Rheumatol*. 2015;33(4 Suppl 91):S117-S122.
90. Corazza GR, Menozzi MG, Strocchi A, et al. The diagnosis of small bowel bacterial overgrowth. Reliability of jejunal culture and

- inadequacy of breath hydrogen testing. *Gastroenterology*. 1990;98(2):302-309.
91. Neale G, Gompertz D, Schonsby H, Tabaqchali S, Booth CC. The metabolic and nutritional consequences of bacterial overgrowth in the small intestine. *Am J Clin Nutr*. 1972;25(12):1409-1417.
 92. Jimenez L, Stamm DA, Depaula B, Duggan CP. Is serum methylmalonic acid a reliable biomarker of vitamin B12 status in children with short bowel syndrome: a case series. *J Pediatr*. 2018;192:259-261.
 93. Piper HG, Fan D, Coughlin LA, et al. Severe gut microbiota dysbiosis is associated with poor growth in patients with short bowel syndrome. *JPEN J Parenter Enteral Nutr*. 2017;41(7):1202-1212.
 94. Mayeur C, Gratadoux JJ, Bridonneau C, et al. Faecal D/L lactate ratio is a metabolic signature of microbiota imbalance in patients with short bowel syndrome. *PLoS One*. 2013;8(1):e54335.
 95. Berean KJ, Ha N, Ou JZ, et al. The safety and sensitivity of a telemetric capsule to monitor gastrointestinal hydrogen production in vivo in healthy subjects: a pilot trial comparison to concurrent breath analysis. *Aliment Pharmacol Ther*. 2018;48(6):646-654.
 96. Pimentel M, Saad RJ, Long MD, Rao SSC. ACG clinical guideline: small intestinal bacterial overgrowth. *Am J Gastroenterol*. 2020;115(2):165-178.
 97. Quigley EMM, Murray JA, Pimentel M. AGA clinical practice update on small intestinal bacterial overgrowth: expert review. *Gastroenterology*. 2020;159(4):1526-1532.
 98. Mutanen A, Lohi J, Heikkilä P, Koivusalo AI, Rintala RJ, Pakarinen MP. Persistent abnormal liver fibrosis after weaning off parenteral nutrition in pediatric intestinal failure. *Hepatology*. 2013;58(2):729-738.
 99. Colomb V, Dabbas-Tyan M, Taupin P, et al. Long-term outcome of children receiving home parenteral nutrition: a 20-year single-center experience in 302 patients. *J Pediatr Gastroenterol Nutr*. 2007;44(3):347-353.
 100. Hess RA, Welch KB, Brown PI, Teitelbaum DH. Survival outcomes of pediatric intestinal failure patients: analysis of factors contributing to improved survival over the past two decades. *J Surg Res*. 2011;170(1):27-31.
 101. Lloyd DA, Vega R, Bassett P, Forbes A, Gabe SM. Survival and dependence on home parenteral nutrition: experience over a 25-year period in a UK referral centre. *Aliment Pharmacol Ther*. 2006;24(8):1231-1240.
 102. Cowles RA, Ventura KA, Martinez M, et al. Reversal of intestinal failure-associated liver disease in infants and children on parenteral nutrition: experience with 93 patients at a referral center for intestinal rehabilitation. *J Pediatr Surg*. 2010;45(1):84-87; discussion 87-88.
 103. Diamond IR, de Silva N, Pencharz PB, et al. Neonatal short bowel syndrome outcomes after the establishment of the first Canadian multidisciplinary intestinal rehabilitation program: preliminary experience. *J Pediatr Surg*. 2007;42(5):806-811.
 104. Javid PJ, Malone FR, Reyes J, Healey PJ, Horslen SP. The experience of a regional pediatric intestinal failure program: successful outcomes from intestinal rehabilitation. *Am J Surg*. 2010;199(5):676-679.
 105. Sigalet D, Boctor D, Robertson M, et al. Improved outcomes in paediatric intestinal failure with aggressive prevention of liver disease. *Eur J Pediatr Surg*. 2009;19(6):348-353.
 106. Torres C, Sudan D, Vanderhoof J, et al. Role of an intestinal rehabilitation program in the treatment of advanced intestinal failure. *J Pediatr Gastroenterol Nutr*. 2007;45(2):204-212.
 107. Diamond IR, Sterescu A, Pencharz PB, Kim JH, Wales PW. Changing the paradigm: omegaven for the treatment of liver failure in pediatric short bowel syndrome. *J Pediatr Gastroenterol Nutr*. 2009;48(2):209-215.
 108. Merras-Salmio L, Mutanen A, Ylinen E, Rintala R, Koivusalo A, Pakarinen MP. Pediatric intestinal failure: the key outcomes for the first 100 patients treated in a national tertiary referral center during 1984-2017. *JPEN J Parenter Enteral Nutr*. 2018;42(8):1304-1313.
 109. Venick RS. Long-term results of intestinal transplantation in children: survival after 10 years, intestinal function, quality of life. *Curr Opin Organ Transplant*. 2018;23(2):219-223.
 110. Lal S, Pironi L, Wanten G, et al; Home Artificial Nutrition & Chronic Intestinal Failure Special Interest Group of the European Society for Clinical Nutrition and Metabolism (ESPEN). Clinical approach to the management of Intestinal Failure Associated Liver Disease (IFALD) in adults: a position paper from the Home Artificial Nutrition and Chronic Intestinal Failure Special Interest Group of ESPEN. *Clin Nutr*. 2018;37(6 Pt A):1794-1797.
 111. US Food and Drug Administration Center for Drug Evaluation and Research. Gastroenterology Regulatory Endpoints and the Advancement of Therapeutics (GREAT): Parenteral Nutrition-Induced Liver Disease (PNALD) Workshop; September 19-24, 2012; College Park, MD.
 112. Armuzzi A, Marcocchia S, Zocco MA, et al. Non-Invasive assessment of human hepatic mitochondrial function through the ¹³C-methionine breath test. *Scand J Gastroenterol*. 2000;35(6):650-653.
 113. Dufour DR, Lott JA, Nolte FS, Gretch DR, Koff RS, Seeff LB. Diagnosis and monitoring of hepatic injury. I. Performance characteristics of laboratory tests. *Clin Chem*. 2000;46(12):2027-2049.
 114. McIntyre N. The limitations of conventional liver function tests. *Semin Liver Dis*. 1983;3(4):265-274.
 115. Fitzgibbons SC, Jones BA, Hull MA, et al. Relationship between biopsy-proven parenteral nutrition-associated liver fibrosis and biochemical cholestasis in children with short bowel syndrome. *J Pediatr Surg*. 2010;45(1):95-99; discussion 99.
 116. Woodward JM, Priest AN, Hollingsworth KG, Lomas DJ. Clinical application of magnetic resonance spectroscopy of the liver in patients receiving long-term parenteral nutrition. *JPEN J Parenter Enteral Nutr*. 2009;33(6):669-676.
 117. Pichler J, Horn V, Macdonald S, Hill S. Intestinal failure-associated liver disease in hospitalised children. *Arch Dis Child*. 2012;97(3):211-214.
 118. Kamath PS, Wiesner RH, Malinchoc M, et al. A model to predict survival in patients with end-stage liver disease. *Hepatology*. 2001;33(2):464-470.
 119. McDiarmid SV, Anand R, Lindblad AS; Principal Investigators and Institutions of the Studies of Pediatric Liver Transplantation (SPLIT) Research Group. Development of a pediatric end-stage liver disease score to predict poor outcome in children awaiting liver transplantation. *Transplantation*. 2002;74(2):173-181.
 120. Duro D, Fitzgibbons S, Valim C, et al. [¹³C]Methionine breath test to assess intestinal failure-associated liver disease. *Pediatr Res*. 2010;68(4):349-354.
 121. Hong CR, Han SM, Staffa SJ, Carey AN, Lee CK, Modi BP. Noninvasive assessment of liver fibrosis in pediatric intestinal failure patients using liver stiffness measurement by vibration-controlled transient elastography. *J Pediatr Surg*. 2019;54(6):1174-1178.
 122. Lodwick D, Dienhart M, Cooper JN, et al. A pilot study of ultrasound elastography as a non-invasive method to monitor liver disease in children with short bowel syndrome. *J Pediatr Surg*. 2017;52(6):962-965.
 123. Van Gossum A, Pironi L, Messing B, et al. Transient Elastography (FibroScan) is not correlated with liver fibrosis but with cholestasis in patients with long-term home parenteral nutrition. *JPEN J Parenter Enteral Nutr*. 2015;39(6):719-724.
 124. Lee WS, Sokol RJ. Intestinal microbiota, lipids, and the pathogenesis of intestinal failure-associated liver disease. *J Pediatr*. 2015;167(3):519-526.
 125. Khalaf RT, Sokol RJ. New insights into intestinal failure-associated liver disease in children. *Hepatology*. 2020;71(4):1486-1498.
 126. Courtney CM, Warner BW. Pediatric intestinal failure-associated liver disease. *Curr Opin Pediatr*. 2017;29(3):363-370.

127. Wales PW, de Silva N, Kim JH, Lecce L, Sandhu A, Moore AM. Neonatal short bowel syndrome: a cohort study. *J Pediatr Surg*. 2005;40(5):755-762.
128. Lacaille F, Gupte G, Colomb V, et al; ESPGHAN Working Group of Intestinal Failure and Intestinal Transplantation. Intestinal failure-associated liver disease: a position paper of the ESPGHAN working group of intestinal failure and intestinal transplantation. *J Pediatr Gastroenterol Nutr*. 2015;60(2):272-283.
129. Nasr A, Diamond IR, de Silva NT, Wales PW. Is the use of parenteral omega-3 lipid emulsions justified in surgical neonates with mild parenteral nutrition-associated liver dysfunction?. *J Pediatr Surg*. 2010;45(5):980-986.
130. Christensen RD, Henry E, Wiedmeier SE, Burnett J, Lambert DK. Identifying patients, on the first day of life, at high-risk of developing parenteral nutrition-associated liver disease. *J Perinatol*. 2007;27(5):284-290.
131. Buchman AL, Naini BV, Spilker B. The differentiation of intestinal-failure-associated liver disease from nonalcoholic fatty liver and non-alcoholic steatohepatitis. *Semin Liver Dis*. 2017;37(1):33-44.
132. Diamond IR, de Silva NT, Tomlinson GA, et al. The role of parenteral lipids in the development of advanced intestinal failure-associated liver disease in infants: a multiple-variable analysis. *JPEN J Parenter Enteral Nutr*. 2011;35(5):596-602.
133. Diamond IR, Grant RC, Pencharz PB, et al. Preventing the progression of intestinal failure-associated liver disease in infants using a composite lipid emulsion: a pilot randomized controlled trial of SMOFlipid. *JPEN J Parenter Enteral Nutr*. 2017;41(5):866-877.
134. Willis TC, Carter BA, Rogers SP, Hawthorne KM, Hicks PD, Abrams SA. High rates of mortality and morbidity occur in infants with parenteral nutrition-associated cholestasis. *JPEN J Parenter Enteral Nutr*. 2010;34(1):32-37.
135. Mueller S, Sandrin L. Liver stiffness: a novel parameter for the diagnosis of liver disease. *Hepat Med*. 2010;2:49-67.
136. Huijbers A, Koggel LM, Bronkhorst C, Verheij J, Wanten GJA. Systematic review: noninvasive assessments of intestinal failure-associated liver disease in the adult population. *JPEN J Parenter Enteral Nutr*. 2019;43(5):615-626.
137. Modi BP, Langer M, Ching YA, et al. Improved survival in a multidisciplinary short bowel syndrome program. *J Pediatr Surg*. 2008;43(1):20-24.
138. Nucci A, Burns RC, Armah T, et al. Interdisciplinary management of pediatric intestinal failure: a 10-year review of rehabilitation and transplantation. *J Gastrointest Surg*. 2008;12(3):429-435; discussion 435-426.
139. Javid PJ, Wendel D, Horslen SP. Organization and outcomes of multidisciplinary intestinal failure teams. *Semin Pediatr Surg*. 2018;27(4):218-222.
140. Jones BA, Hull MA, McGuire MM, Kim HB. Autologous intestinal reconstruction surgery. *Semin Pediatr Surg*. 2010;19(1):59-67.
141. Modi BP, Javid PJ, Jaksic T, et al. First report of the international serial transverse enteroplasty data registry: indications, efficacy, and complications. *J Am Coll Surg*. 2007;204(3):365-371.
142. Javid PJ, Oron AP, Duggan CP, Squires RH, Horslen SP; Pediatric Intestinal Failure Consortium. The extent of intestinal failure-associated liver disease in patients referred for intestinal rehabilitation is associated with increased mortality: an analysis of the pediatric intestinal failure consortium database. *J Pediatr Surg*. 2018;53(7):1399-1402.
143. Modi BP, Jaksic T. Pediatric intestinal failure and vascular access. *Surg Clin North Am*. 2012;92(3):729-743, x.
144. Sanchez SE, McAteer JP, Goldin AB, Horslen S, Huebner CE, Javid PJ. Health-related quality of life in children with intestinal failure. *J Pediatr Gastroenterol Nutr*. 2013;57(3):330-334.
145. Jones BA, Hull MA, Richardson DS, et al. Efficacy of ethanol locks in reducing central venous catheter infections in pediatric patients with intestinal failure. *J Pediatr Surg*. 2010;45(6):1287-1293.
146. Mezzoff EA, Fei L, Troutt M, Klotz K, Kocoshis SA, Cole CR. Ethanol lock efficacy and associated complications in children with intestinal failure. *JPEN J Parenter Enteral Nutr*. 2016;40(6):815-819.
147. Casson C, Nguyen V, Nayak P, et al. A comparison of Smoflipid(R) and Intralipid(R) in the early management of infants with intestinal failure. *J Pediatr Surg*. 2020;55(1):153-157.
148. Cober MP, Killu G, Brattain A, Welch KB, Kunisaki SM, Teitelbaum DH. Intravenous fat emulsions reduction for patients with parenteral nutrition-associated liver disease. *J Pediatr*. 2012;160(3):421-427.
149. Gura KM, Lee S, Valim C, et al. Safety and efficacy of a fish-oil-based fat emulsion in the treatment of parenteral nutrition-associated liver disease. *Pediatrics*. 2008;121(3):e678-686.

How to cite this article: Modi BP, Galloway DP, Gura K, et al. ASPEN definitions in pediatric intestinal failure. *J Parenter Enteral Nutr*. 2022;46:42-59.
<https://doi.org/10.1002/jpen.2232>